

Power Optimization Testing Exploration

Testing Keys

Table 1: Sensors + Microcontroller - Testing Key - (For sensor variables multimeter is in series with 3.3 V and sensor VIN)

Test ID	Resolution	Ranging Frequency (Hz)	Timing Budget	Mode	WiFi	ESP Sleep	VHV Repeat #
	Independent Variable: Ranging Frequency (Hz)						
1	8x8	Default (1 Hz)	Default (10 ms)	Autonomous	Off	No	Natural default (commented out)
2	8x8	1 Hz	Default (10 ms)	Autonomous	Off	No	Default
3	8x8	2 Hz	Default (10 ms)	Autonomous	Off	No	Default
4	8x8	5 Hz	Default (10 ms)	Autonomous	Off	No	Default
5	8x8	10 Hz	Default (10 ms)	Autonomous	Off	No	Default
6	8x8	20 Hz	Default (10 ms)	Autonomous	Off	No	Default
7	8x8	Default	Default (10 ms)	Autonomous	Off	No	Default
	Test 1 and 7 are meant to be the same, I just want to rerun the initial (default) test after the frequency sweep occurs.						
	Independent Variable: Resolution - (Test 9b is a confirmation test of sensor behavior)						
8	8x8	1 Hz	Default (10 ms)	Autonomous	Off	No	Default
9a	4x4	1 Hz	Default (10 ms)	Autonomous	Off	No	Default
9b	4x4	10 Hz	Default (10 ms)	Autonomous	Off	No	Default
	Independent Variable: Timing Budget (integration)						
10a	8x8	10 Hz (using 10 instead of 1 Hz because 1Hz gives the multimeter little chance of capturing active current)	Reduced (2 ms)	Autonomous	Off	No	Default
10b	8x8	10 Hz	Reduced (3 ms)	Autonomous	Off	No	Default
10c	8x8	10 Hz	Reduced (5 ms)	Autonomous	Off	No	Default
10d	8x8	10 Hz	Reduced (7 ms)	Autonomous	Off	No	Default
	Independent Variable: Mode						
11	8x8	1 Hz	Default (10 ms)	Continuous	Off	No	Default
	Independent Variable: WiFi						
12	8x8	1 Hz	Default (10 ms)	Autonomous	On	No	Default
	Independent Variable: Sleep						
13	8x8	1 Hz	Default (10 ms)	Autonomous	Off	Yes	Default
	Sleep being yes means ESP32 Light sleep is enabled between ranging cycles						
	Independent Variable: VHV Repeat Count						
14	8x8	10 Hz (using 10 instead of 1 Hz because 1Hz gives the multimeter little chance of capturing active current)	Default (10 ms)	Autonomous	Off	No	1
15	8x8	10 Hz	Default (10 ms)	Autonomous	Off	No	10
16	8x8	10 Hz	Default (10 ms)	Autonomous	Off	No	40
17	8x8	10 Hz	Default (10 ms)	Autonomous	Off	No	255

Table 2: Buzzers - Testing Key

Test ID	Buzzer Setting
1	Buzzer (Continuous)
2	Buzzer (Pulsed)

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Results

Table 3: Cold Boot Startup - Results

Test	Min Current (mA)	Avg Current (mA)	Max Current (mA)	Notes
Cold Boot Startup	20	50	80	Both the minimum and maximum values were quick flashes resulting in an approximation of the average as a perfect middle value.

Table 4: Sensors + Microcontroller - Results

Test ID	Min Current - Sensor (mA)	Min Current - System (mA)	Max Current - Sensor (mA)	Max Current - System (mA)	Avg Current - Sensor (mA)	System Total Avg (mA)	Notes	Calculated Avg Sensor Current (mA)
1 (Default)	4.209	Not sure	4.228	70	4.215	~ 50	Additionally, the variable Target Order is embedded. This is outlined in the Ideas tab of this google file. For default it is set to STRONGEST .	~4.6
2	Ends up being same as test 1 (default) and test 7 because 1 Hz is the default ranging frequency							
3	42.27	30	42.39	60	42.25	~40	Multimeter is recording active current measurement rather than average. Active Current: ~42mA	~4.96
4	42.34	30	42.70	50	42.30	~40	Active Current: ~42mA	~6.09
5	42.84	50	44.54	80	43.60	~60	Active Current: ~43mA	~7.98
6	66.65	50	67.62	130	67.35	Betw. 80-120 → ~100	20 Hz exceeds the documented ULD limit of 15 Hz for 8x8 resolution leading to unexpected sensor behavior	~11.76
7	4.205	40	4.345	70	4.225	~50	The most likely reason the current is so low is b/c the multimeter is catching the sensor in its Idle state due to the sensor only sampling once per second. From this we can assume the sensor's idle current draw is ~4.2 mA	~4.58
8	4.232	30	4.236	70	4.235	~50	Same test as test 1 and 7.	~4.61
9a	0.311 (Idle Current)	30	8.121 (Partial capture of active current)	50	The average is bouncing between 0.311 and around 7.2 - 7.9 mA.	~40	Multimeter captures a mix of idle (~0.311 mA) and partial active states at 1 Hz due to 1% duty cycle. True active current confirmed as ~15.7 mA from Test 9b at 10 Hz characterization. Max reading of 8.121 mA represents partial capture of active period rather than true peak. Idle current is significantly lower than 8x8 (~0.311 mA vs ~4.2 mA), suggesting the sensor enters a deeper idle state at 4x4 resolution.	~0.465
9b	15.292	40	16.360	60	~15.7	~50	Additional characterization measurement performed at 10 Hz with 4x4 to establish true active current. Result: ~15.7 mA active current. (Only caught active current here, sampling was too frequent to get the idle current as the low current value) Thus we assume the idle when changing from 8x8 to 4x4 is 0.311 from Test 9a.	~1.85
10a	24.320	40	25.263	70	~24.73	~60	We use 10 Hz here to maximize our chances of capturing active current. We then use the duty cycle calculation at 1 Hz for each of these active currents to determine the average current with our default sampling frequency value (1 Hz) for each trial. It is apparent that the true average at 1 Hz barely changes across all integration times — only 0.36 mA difference (from default test 1 (4.6 mA) and test 10a (4.24 mA)) between 2 ms and 10 ms. This is because at 1 Hz the duty cycle is so small that idle current dominates the average regardless of how long the integration window is. Thus, reducing integration time at 1 Hz gives negligible power savings at the average current level.	~4.24
10b	25.250	40	25.252	60	~25.251	~60		~4.27
10c	30.656	60	31.098	70	~30.932	~65		~4.33
10d	36.08	50	36.90	80	~36.58	~60		~4.46
11	65.94	80	66.93	130	~66.42	~85	In CONTINUOUS mode the sensor integrates for the full 1000 ms frame period at 1 Hz. The sensor draws a sustained ~66.42 mA with no idle periods, compared to ~4.58 mA calculated average in AUTONOMOUS mode — a ~93% reduction. The higher sustained current (~66 mA) vs autonomous burst current (~42 mA) confirms that	~66.42

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Test ID	Min Current - Sensor (mA)	Min Current - System (mA)	Max Current - Sensor (mA)	Max Current - System (mA)	Avg Current - Sensor (mA)	System Total Avg (mA)	Notes	Calculated Avg Sensor Current (mA)
							AUTONOMOUS mode not only duty cycles the sensor but also reduces current during the active burst itself through internal power gating.	
12	Not yet at the stage to measure WiFi's effects							
13	4.222	5	4.240	75	~4.228	~6.0	Test 13 implements ESP32 light sleep between ranging cycles using the sensor's INT (interrupt) pin connected to GPIO 5. The sensor ranges continuously and autonomously at 1 Hz while the ESP32 sleeps between measurements. When the sensor completes a measurement and data is ready, it pulses the INT pin LOW which immediately wakes the ESP32. The ESP32 reads the data via I2C, then returns to light sleep until the next INT pulse. This minimizes ESP32 active time to only the brief window needed for I2C data transfer (~5-10 ms per cycle). Results confirmed this approach is highly effective – the sensor current remained steady at ~4.238 mA (identical to Test 1 baseline) while the DC Power supply consistently read 5mA with short bursts to ~40-80mA indicating the ESP32 light sleep current is negligibly small. When the USB is attached as the source of power from the power bank, the avg system current draw is ~11mA. Thus, the USB-serial chip and dev board circuitry add ~6mA.	~4.58
14	53.30	73	60.59	113	~56.68	~105	VHV set to 1 (every frame). Higher current than natural default (~44 mA) suggests explicit VHV=1 adds calibration overhead vs uninitialised default.	~9.45
15	45.05	73	51.24	101	~47.87	~98	Measureable reduction from Test 14	~6.57
16	44.46	73	51.34	102	~45.29	~94	Diminishing returns vs. Test 15	~6.31
17	44.36	73	50.22	102	~45.15	~92	VHV set to 255 (calibration at initialization only) near identical results to Test 16	~6.30

Sample Calculations:

- **Calculated Avg Current - Test 4 (5 Hz Ranging Frequency)**
 - $Duty\ Cycle = Integration\ time / Frame\ period \rightarrow 10\ ms / 200\ ms \rightarrow = 0.05\ (5\%)$
 - $I_{avg} = (I_{active} \times Duty\ Cycle) + (I_{idle} \times (1 - Duty\ Cycle)) \rightarrow (42\ mA \times 0.05) + (4.2\ mA \times 0.95) \rightarrow = 6.09\ mA$
 - Explanation: The sensor spends 5% of its time drawing 42 mA and 95% of its time drawing 4.2 mA.

Notes:

- While the measurements are not exactly the values we wanted (average current, min current, etc.) based off the situation (the multimeter's limited sampling ability) and the variance in measured values we can infer that we measured idle current in test 1 rather than average current, and active current for tests 3, 4, and 5 rather than average current.
- **Test 1/7:** True active current estimated at ~42 mA based on Tests 3-5.
 - At 1 Hz with 10 ms integration the duty cycle is ~1%, causing the multimeter to predominantly sample the idle state rather than the active state. True average current at 1 Hz is calculated as approximately 4.6 mA using the duty cycle formula rather than direct measurement.
- **Test 3/4:** Similar readings suggest multimeter consistently capturing active current at these frequencies
- **Test 6:** 20 Hz exceeds the documented ULD limit of 15 Hz for 8x8 resolution. The significantly higher current draw (~67 mA vs ~42 mA at lower frequencies) suggests unexpected sensor behavior beyond the rated operating limit. This data point should not be used for power optimization decisions and is included for reference only.

Table 5: Buzzer - Results

Test ID	Min Current (mA)	Avg Current (mA)	Max Current (mA)	System Total Avg (mA)	Notes	Calculated avg (mA)
1	2.09	~2.156	2.235	71	Continuous alarm, thus results should be similar from min, max, and avg current.	N/A
2	0	Jumping between 0 and 2.16.	~2.16	Minimum: 36 Maximum: 41 Average: 37	If the buzzer is active for 20% of the time then we can do a duty cycle calculation for the avg current draw.	~ 0.432

Overall notes: Buzzer characterization was performed using an SFM-27 active buzzer powered at 3.3V. Two tests were conducted: B1 (continuous) and B2 (pulsed at 20% duty cycle, 200ms on / 800ms off). The multimeter, placed in series between the 3.3V supply and the buzzer positive terminal, consistently measured ~2.1mA active current. This was initially questioned due to a 35mA system total jump observed on the bench supply when the buzzer activated, however isolated testing with the buzzer connected directly to the bench supply (no ESP32) confirmed the true buzzer current is ~2.1mA at 3.3V. The 35mA system jump was attributed to ESP32 internal behavior changes when toggling

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GPIO 1 rather than the buzzer itself. At 20% duty cycle (representative of a proximity alert scenario where the buzzer beeps when an object is within 150cm), the calculated average buzzer current is ~0.42mA, making it a negligible contributor to overall system power consumption. The buzzer was confirmed loud and operational at 3.3V throughout testing.

Overall System’s Energy Consumption - Results

Table 6: Baseline (Default) - Energy Consumption Test Results

Test Mode	Duration (minutes)	Min Current Drop (mA)	Max Current Spike (mA)	mAh Consumed	mWh Consumed	Estimated 10hr Runtime (mAh)	Estimated 10hr Runtime (mWh)
Realistic (Using Test ID parameters with beeping buzzer logic)	30	20	70	24	125	480	2500

Table 7: Optimized Configuration - Energy Consumption Test Results

Test Mode	Duration (minutes)	Min Current Drop (mA)	Max Current Spike (mA)	mAh Consumed	mWh Consumed	Estimated 10hr Runtime (mAh)	Estimated 10hr Runtime (mWh)
Optimal Parameters (IDLE state)	30	1	1	0.5	—	10	—
Optimal Parameters (ACTIVE state)	30	5	46	2.75	—	55	—
Optimal Parameters (REALISTIC state)	30	—	—	—	—	~19	—

Notes: Test 10 was conducted using the benchtop DC power supply as the sole power source, connected directly to the ESP32-S3 5V header pin with no USB cable attached. This is because the current draw of the system was too low to keep the USB power bank running. Current readings were read directly from the supply display and mAh values were calculated manually using the formula mAh = avg current (mA) × time (hr). Idle current (no object present) measured a steady ~1 mA, with active current (object within 200 cm, slow beep profile) reading ~5.5 mA average with brief spikes to 26–40 mA on ESP32 wake events. This represents the most accurate reflection of true battery-powered prototype current draw, as no USB circuitry will be present in the final scissor lift deployment. Weighted average at 20% detection rate yields ~0.95 mAh over 30 minutes and an estimated 19 mAh over a 10 hour period, representing approximately a 96% reduction from the Table 6 baseline of 480 mAh. Additionally, the wattage consumed columns are blank due to the DC benchtop supply being unable to provide power readings.

Notes:

- For timing budget / integration times it should be noted that when in CONTINUOUS mode, the default is not one consistent variable for all tests, rather it fluctuates with the specific frequency used during that test. However, AUTONOMOUS mode’s default value is 10 ms.
 - For test 10a - 10d, there are specific integration times that are used, these are noted in the testing key.
- For VHV Repeat Count, the default is 1 calibration every frame.
 - For test 14, 10 represents 1 calibration every 10 frames
 - For test 16, 255 represents only one calibration for the whole operation time, this being at initialization.

Table 8: Test ID + Corresponding Code File

Test ID	Code File	Notes
1 (Default)	Test1_Default_Baseline	N/A
2 - 6	Test2_FrequencySweep	N/A
7	Test1_Default_Baseline	N/A
8, 9a, & 9b	Test3_Resolution	N/A
10a - 10d	Test4_IntegrationTime	Uses AUTONOMOUS mode with specific integration times. The default value used in all other trials is 10 ms.
11	Test5_RangingMode	We set this CONTINUOUS, what continuous mode does is it causes the sensor’s integration time to be the full frame period. For instance, at 1 Hz sampling frequency the duty cycle is approximately 100% (1000ms out of 1000ms). In autonomous mode at 1Hz, the duty cycle is approximately 1%, (10ms out of 1000ms).
12	No file yet	Currently not being tested
13	Test6_ESPPowerModes	T06 measures ESP sleep current floors, providing the minimum possible system current draw when in a light sleep state while sensor is not ranging.

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Test ID	Code File	Notes
1 (Default)	Test1_Default_Baseline	N/A
2 - 6	Test2_FrequencySweep	N/A
14 - 17	Test8_VHVRepeatCount	14 - Calibrate every frame, 15 - Calibrate every 10th frame, 16 - Calibrate every 40th frame, 17 - Calibrate ONCE at initialization

Possible Future Testing:

Table 9: Hardware Shutdown (LPn / XSHUT) - Wake Characterization - Testing Key

Test ID	Sensor State Between Cycles	ESP32 State	Ranging Frequency (Hz)	Resolution	Timing Budget	Re-Init Required	VHV Repeat #
X1	LPn low (Full Shutdown)	Light Sleep	1	4x4	Default	Yes	10
X2	LPn low (Full Shutdown)	Light Sleep	5	4x4	Default	Yes	10
X3	LPn low (Full Shutdown)	Light Sleep	1	8x8	Default	Yes	10
X4	LPn low (Full Shutdown)	Light Sleep	5	8x8	Default	Yes	10

Table 10: Wake Latency + Detection Timing - Results

Test ID	Mode	Ranging Frequency (Hz)	Avg Current (mA)	Wake Latency Avg (ms)	Wake Latency Max (ms)	Detection Delay Avg (ms)	Detection Delay Max (ms)	Notes
W1	Continuous - No sleep	5						
W2	ALP - ESP32 Light Sleep	1						
W3	ALP - ESP32 Light Sleep	2						
W4	ALP - ESP32 Light Sleep	5						
W5	ALP - ESP32 Light Sleep	10						

Table 11: Hardware Shutdown (LPn / XSHUT) - Wake Characterization - Results

Test ID	Shutdown Current (mA)	Wake Peak Current (mA)	Avg Current Over Cycle (mA)	Sensor Boot Time (ms)	Re-init Time (ms)	First Valid Frame Time (ms)	Total Time Wake → Valid (ms)	Notes
X1								
X2								
X3								
X4								